

Technology Stack

Date	25 June 2026
Team ID	SWTID-2026-1236
Project Name	Credit Card Approval Eligibility
Maximum Marks	2 Marks

Technology Stack:

The Technology Stack describes the technologies used to build the Credit Card Approval System. It includes the frontend, backend, database, machine learning libraries, deployment platform, and development tools. These technologies work together to create a web application that collects customer details, predicts credit card approval using AI & ML, and displays the result to the customer quickly.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	• HTML • CSS • JavaScript	Used to build a responsive, user-friendly web interface for customers to fill out the credit card application form.

2	Backend / Server-Side	Python (Flask)	Handles form submission, communicates with the AI & ML model, and returns the prediction result to the website.
3	Cloud / Hosting / Deployment	Render	Used to run and deploy the web application for testing and online access.
4	Version Control & CI/CD	GitHub	Used for version control, code management, and team collaboration.
5	Third-Party APIs / Other Tools	•Pandas •NumPy •Scikit-learn •Joblib •Matplotlib •Seaborn •Jupyter Notebook	• Pandas is used for data manipulation and pre-processing. • NumPy is used for numerical computations. • Scikit-learn is used to build, train, and evaluate the machine learning model. • Joblib is used to save and load the trained model. • Matplotlib and Seaborn are used for data visualization and exploratory data analysis (EDA). • Jupyter Notebook is used for model development, experimentation, and testing.